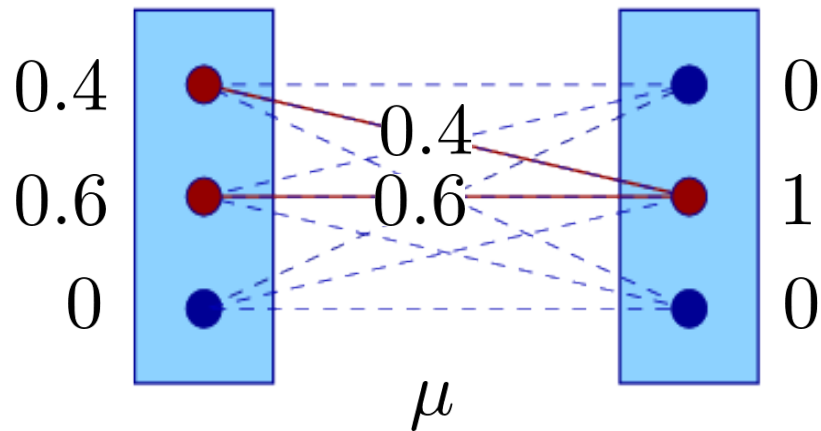
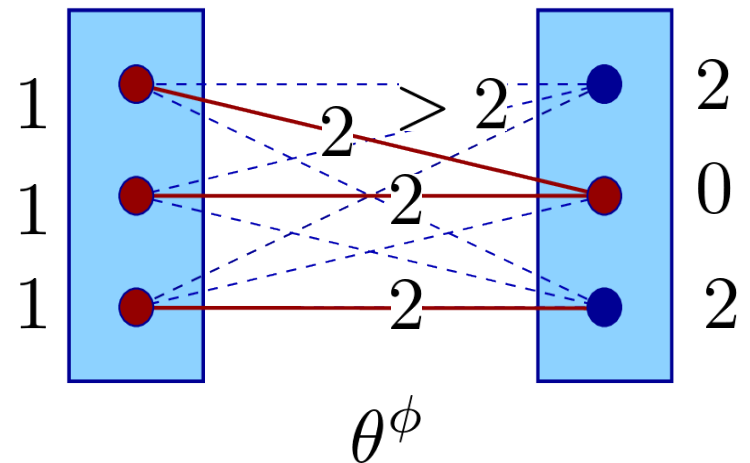


Complementary Slackness

Primal:



Dual:



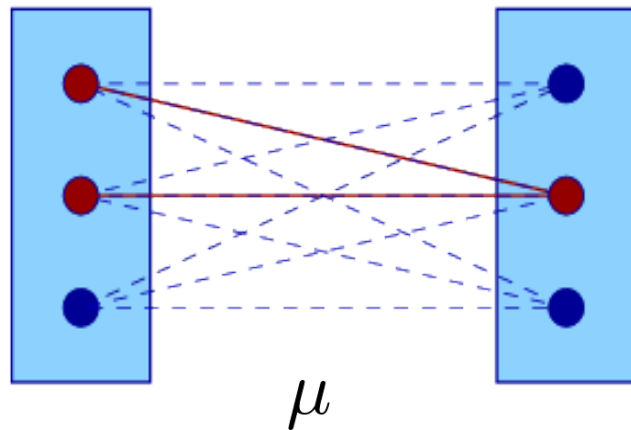
$$\beta_w(s)\mu_w(s) = 0$$

$$(\mu_v(x_v) > 0) \Rightarrow \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

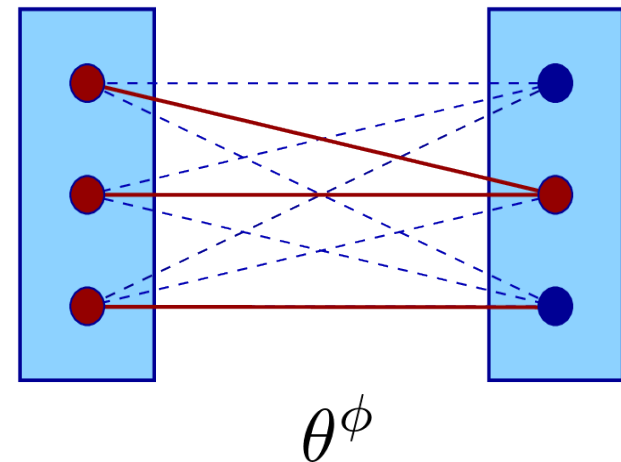
$$(\mu_{uv}(x_u, x_v) > 0) \Rightarrow \theta_v^\phi(x_u, x_v) = \min_{st \in \mathcal{X}_{uv}} \theta_v^\phi(s, t)$$

Recall: Complementary Slackness

Primal:



Dual:



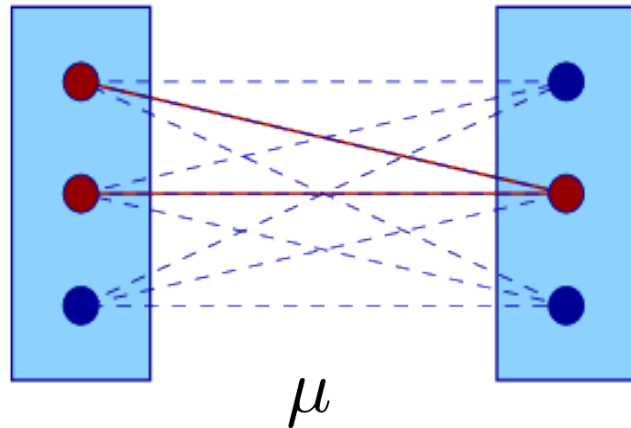
$$\beta_w(s)\mu_w(s) = 0$$

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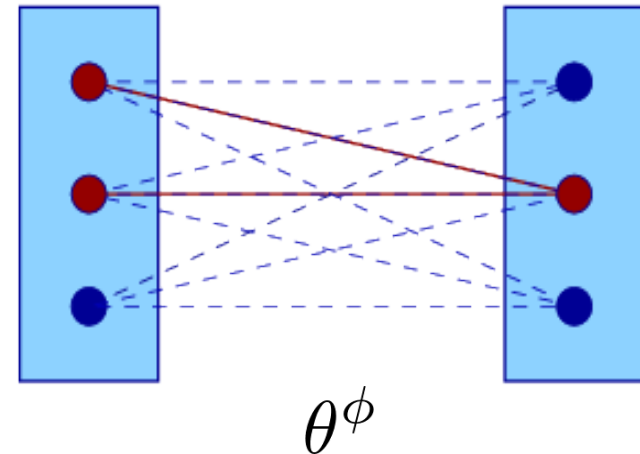
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Recall: Complementary Slackness

Primal:



Dual:



$$\beta_w(s)\mu_w(s) = 0$$

$$(\mu_v(x_v) > 0) \quad \Rightarrow \quad \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

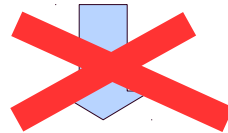
$$(\mu_{uv}(x_u, x_v) > 0) \quad \Rightarrow \quad \theta_v^\phi(x_u, x_v) = \min_{st \in \mathcal{X}_{uv}} \theta_v^\phi(s, t)$$

Dual binarized vector must contain an arc-consistent subset – kernel!

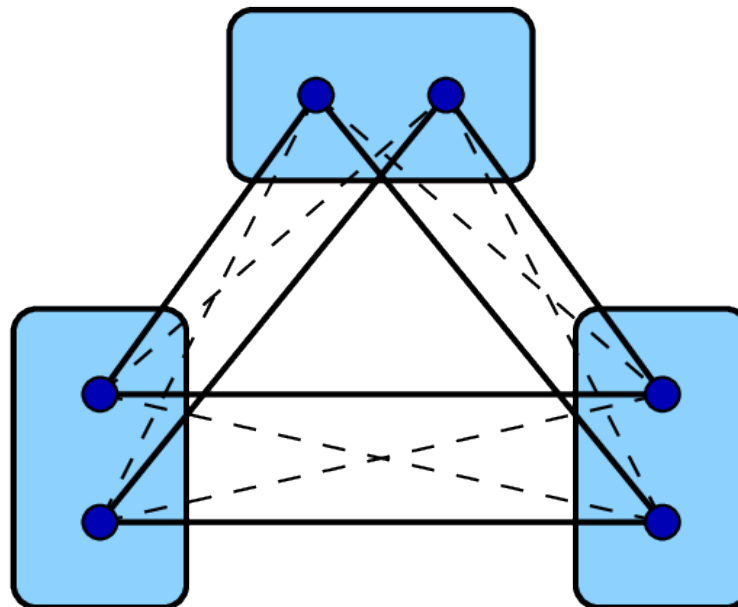
Arc-Consistency vs. (Relaxed) Optimality

Any $\delta(x)$, $x \in \mathcal{X}_v$ is arc-consistent

“non-empty kernel of a dual point”



“there is an integer labeling consisting of red edges and labels”



Arc-Consistency vs. Relaxed Optimality

Any $ce[\mu]$, $\mu \in \mathcal{L}$ is arc-consistent

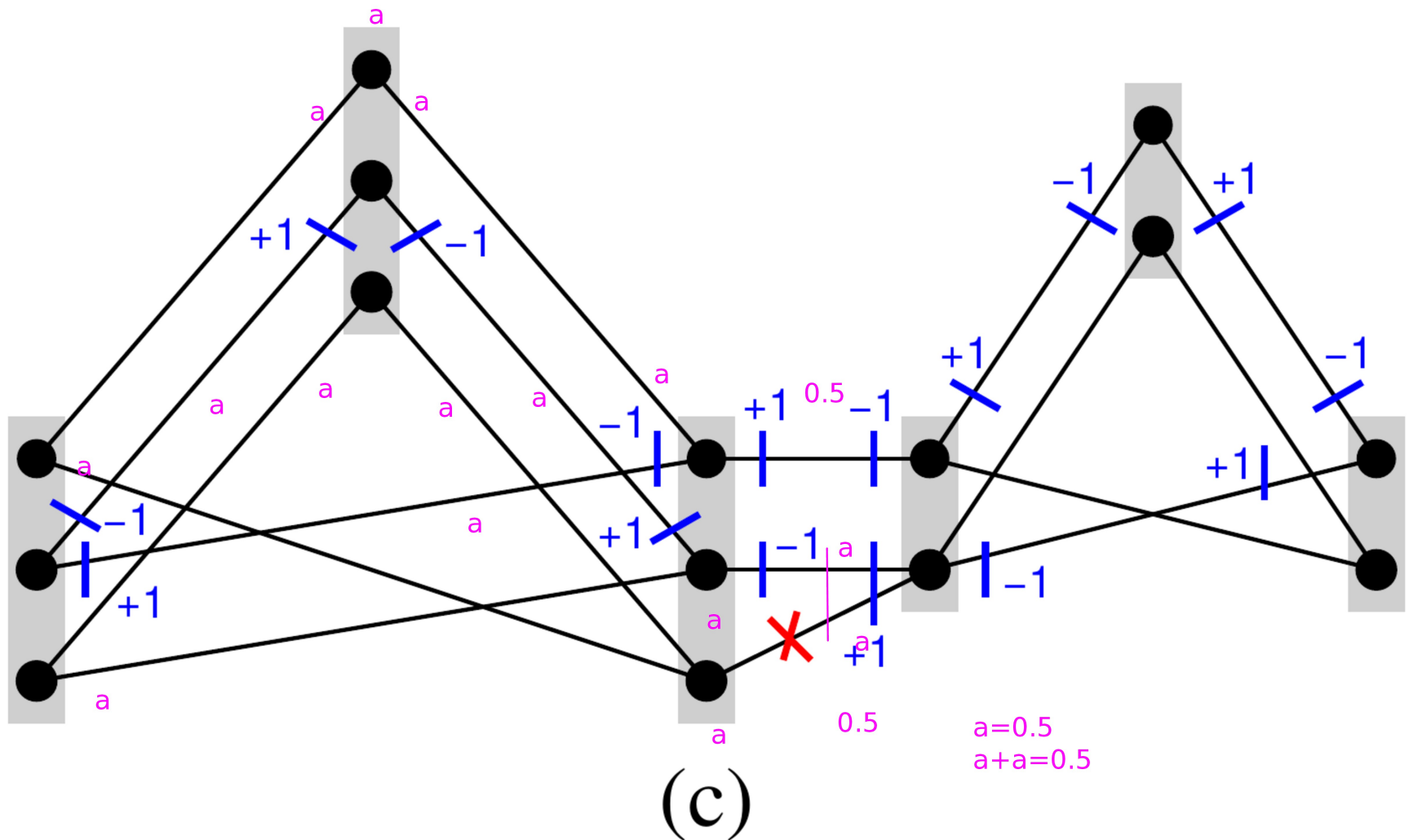
“non-empty kernel of a dual point”



“there is a **relaxed** labeling consisting of red edges and labels”

- 1) Yes
- 2) **No**
- 3) No idea

Arc-Consistency vs. Relaxed Optimality



Picture: Tomas Werner. A Linear Programming Approach to Max-sum Problem: A Review.

Summary: Arc-Consistency vs. (Relaxed) Optimality

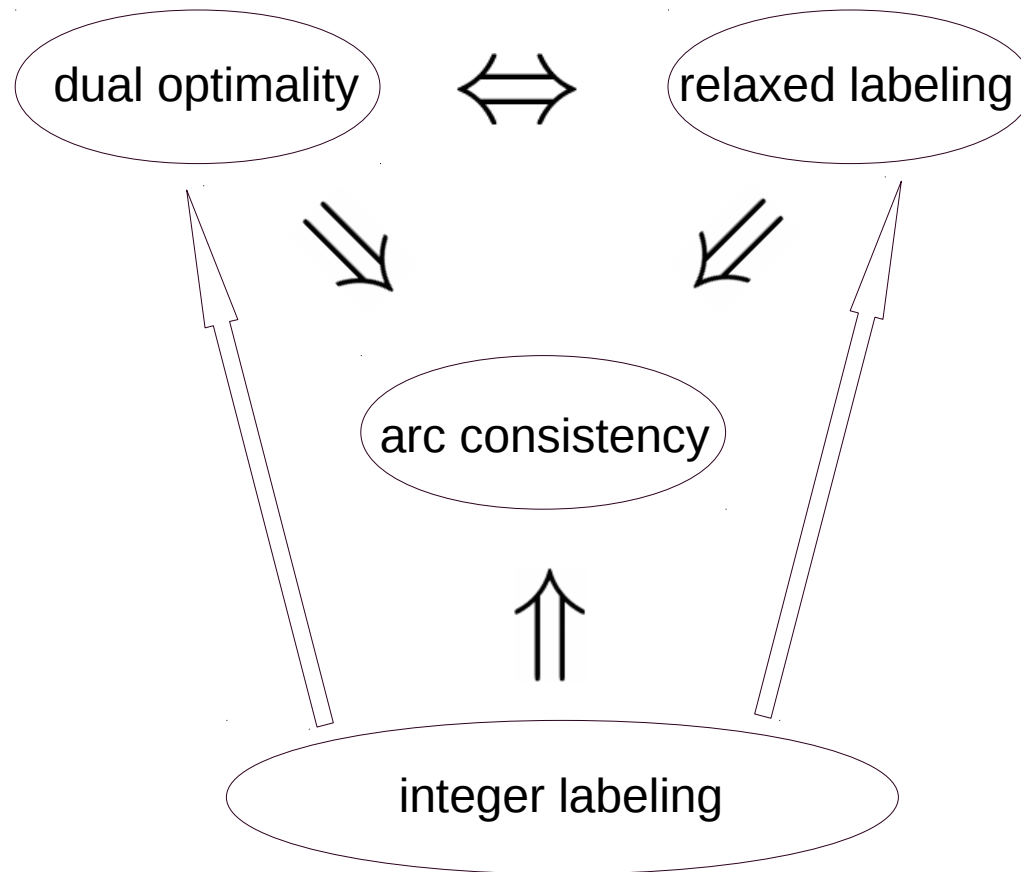
Keywords:

- arc consistency
- dual optimality
- relaxed labeling
- integer labeling

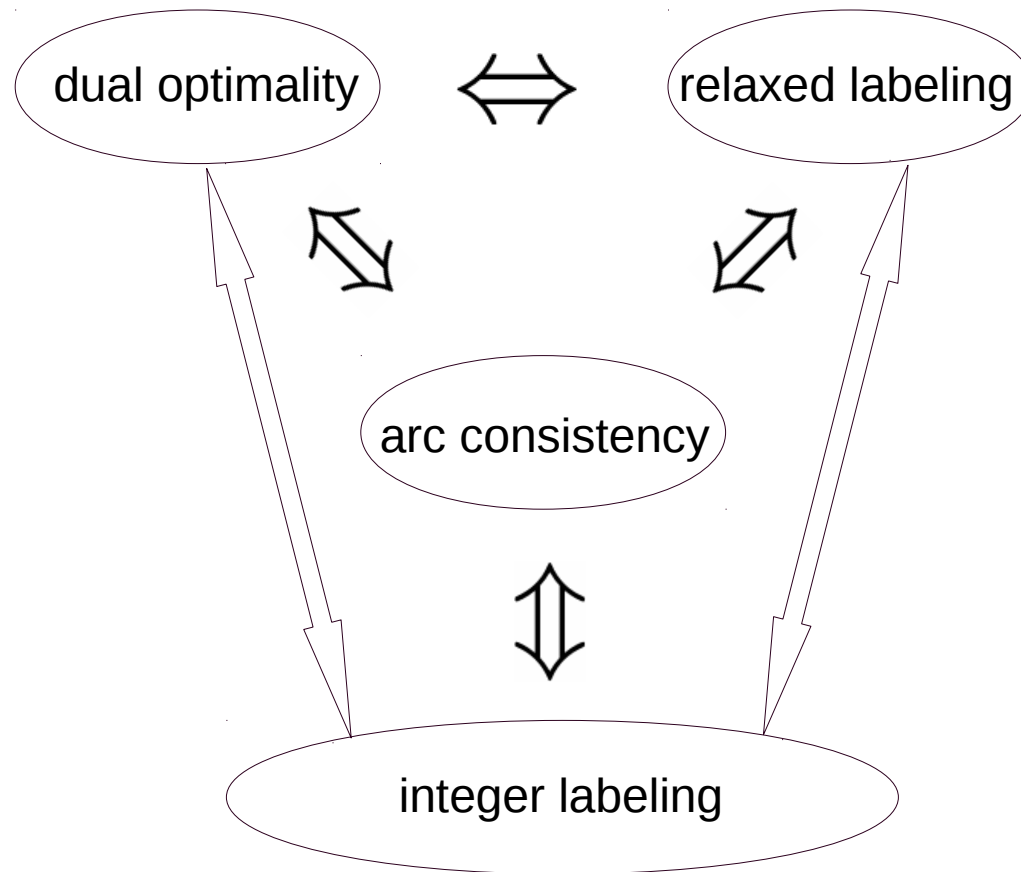
Summary: Arc-Consistency vs. (Relaxed) Optimality

Keywords:

- arc consistency
- dual optimality
- relaxed labeling
- integer labeling



What about... ?



Keywords:

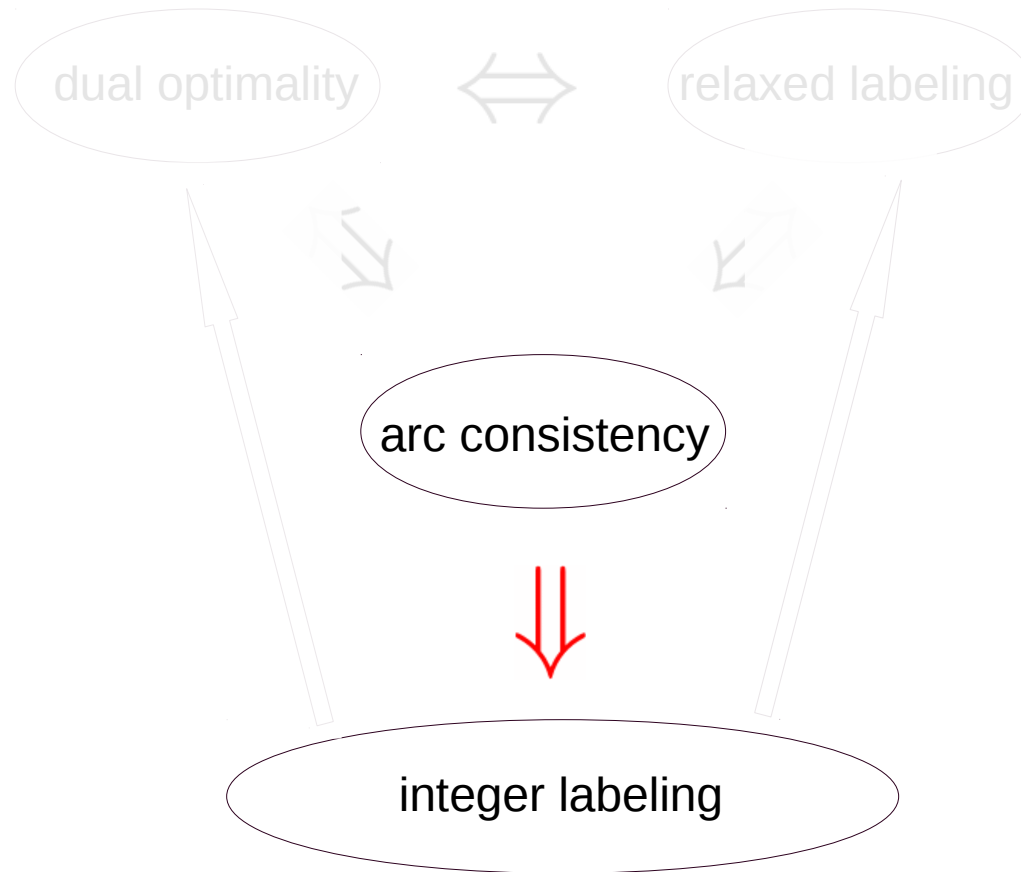
- arc consistency
- dual optimality
- relaxed labeling
- integer labeling

Acyclic models

Proposal: Let G be acyclic and kernel of θ be non-empty.

Then there exists a labeling x consisting of locally minimal edges and labels:

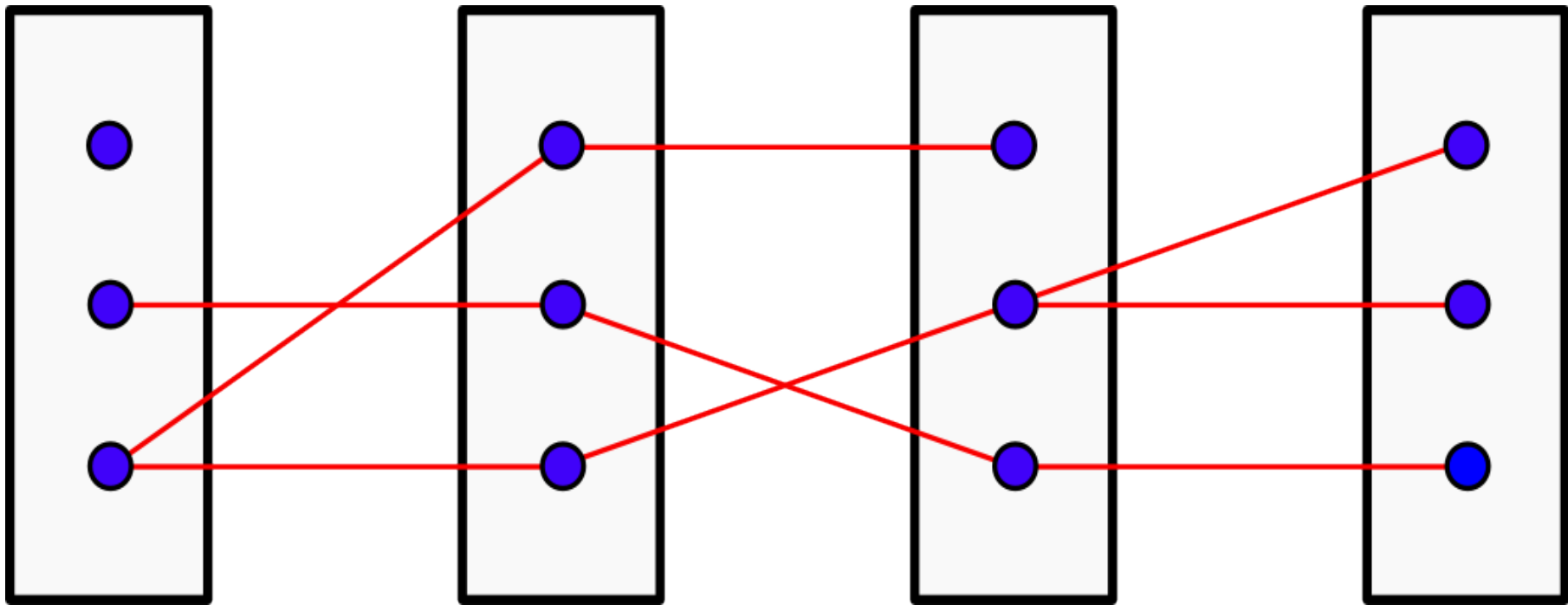
$$\delta(x) \wedge \text{mi}[\theta] = \delta(x)$$

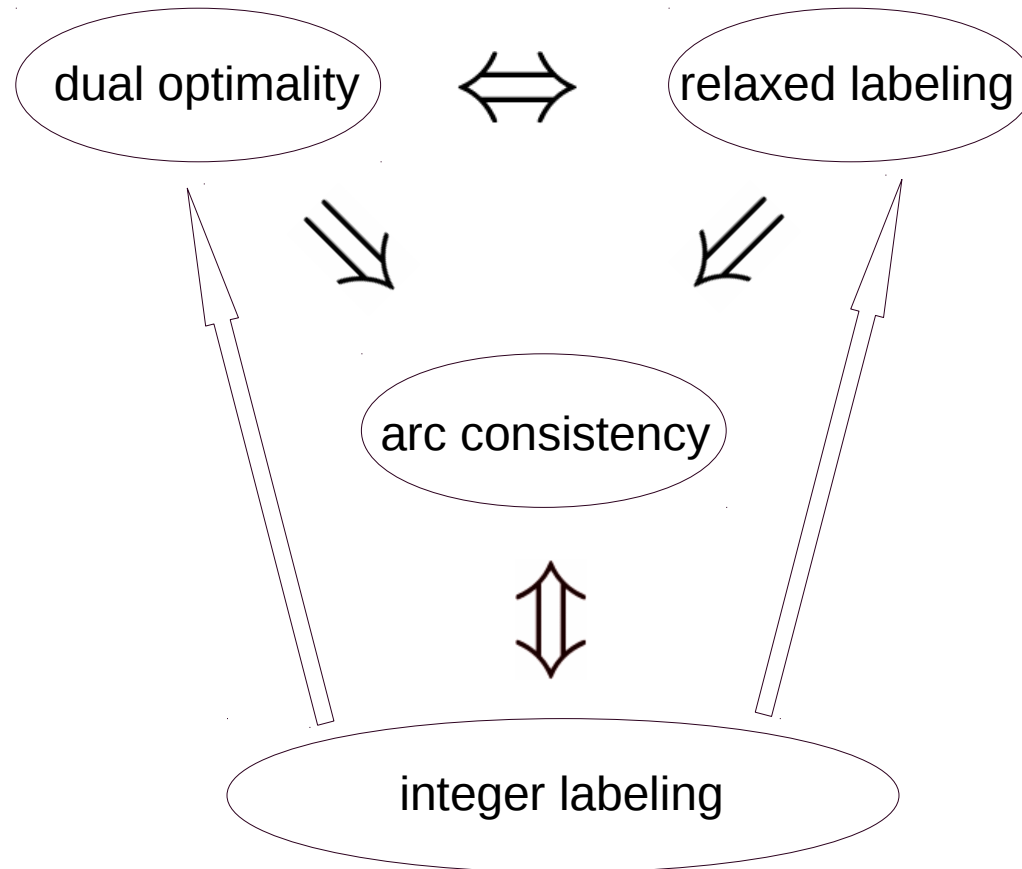


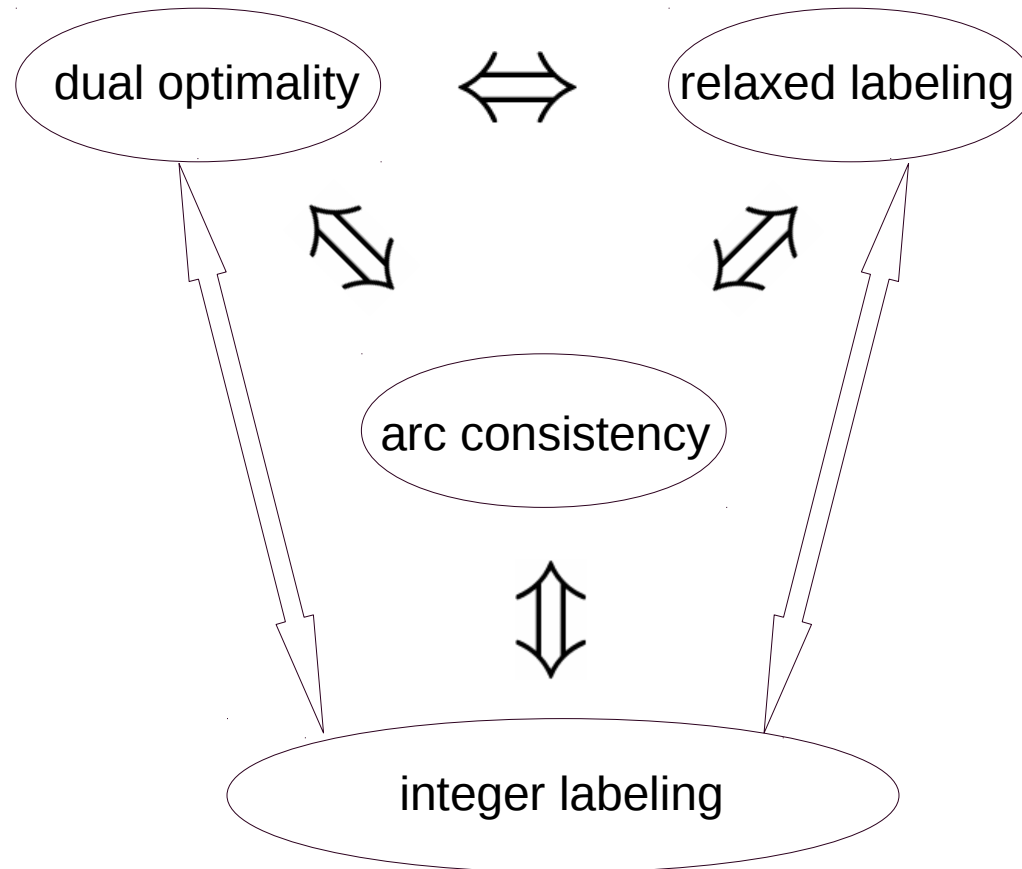
Acyclic models

Proposal: Let G be acyclic and the kernel of $\text{mi}[\theta]$ be non-empty.
Then there exists a labeling x consisting of locally minimal edges and labels:

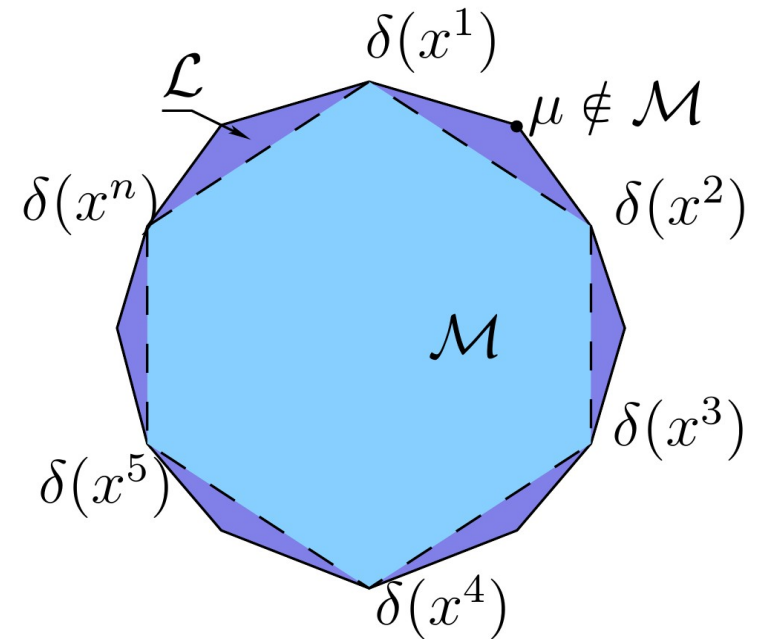
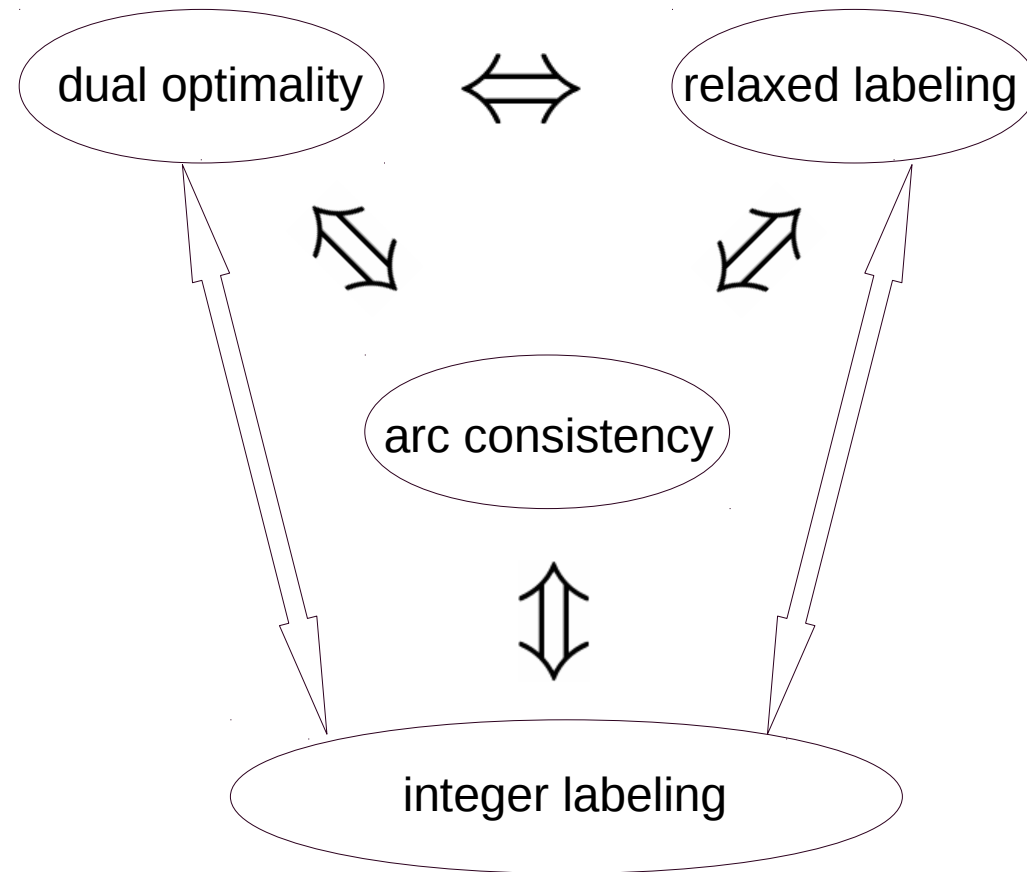
$$\delta(x) \wedge \text{mi}[\theta] = \delta(x)$$







Acyclic models: Local Polytope = Marginal Polytope

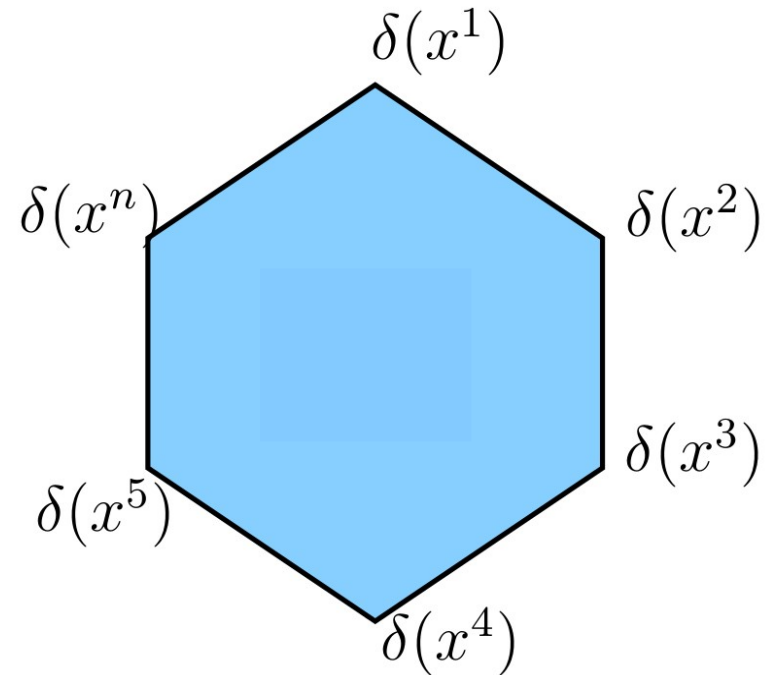
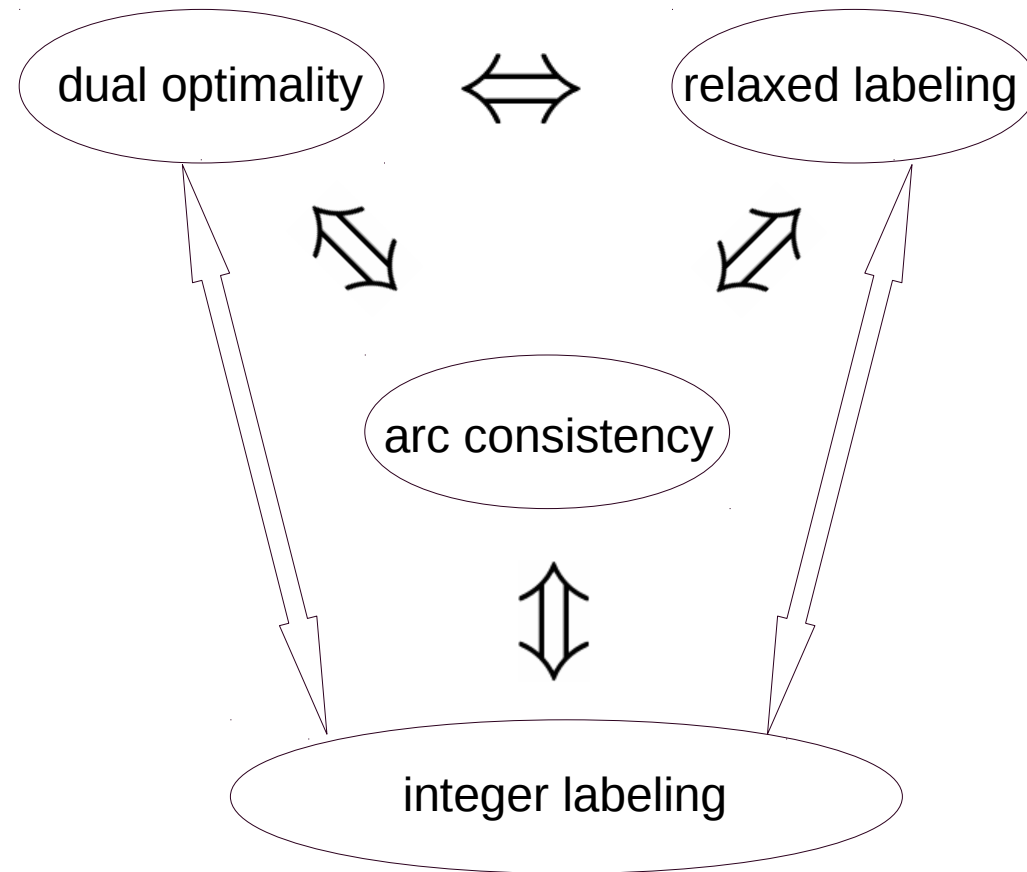


$\mu' \in \mathbb{R}^I$ is a vertex of a polyhedron $\mathcal{L}$ if there exists $\mu \in \mathbb{R}^I$ such that

$$\min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$$

is finite and has a unique solution

Acyclic models: Local Polytope = Marginal Polytope



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